

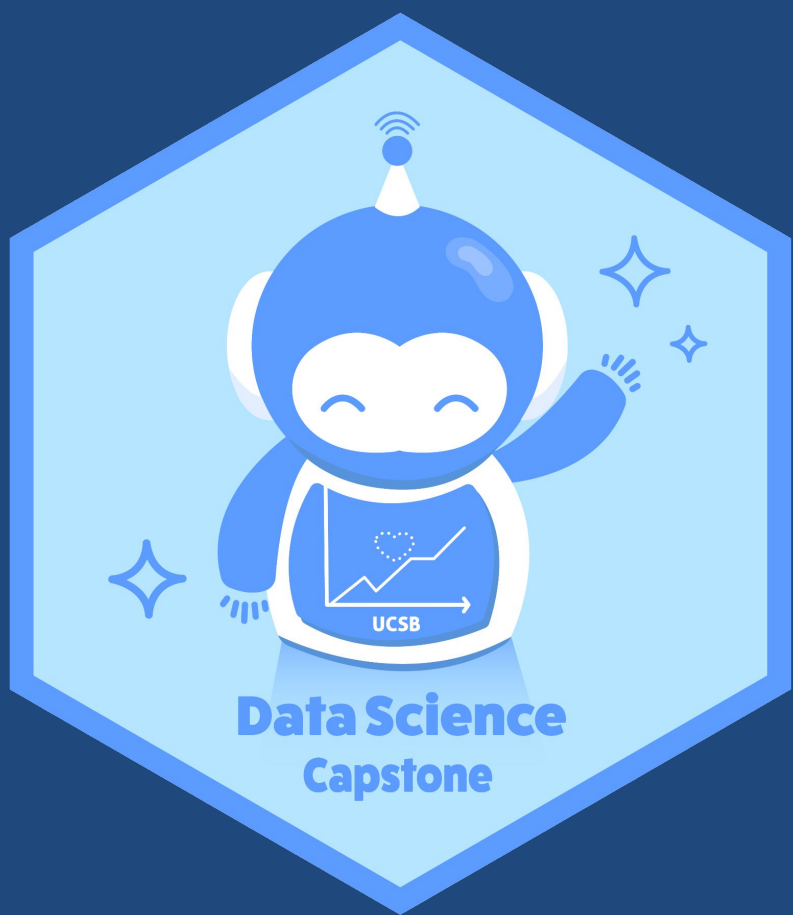
UCSB Geography: Crop Yield Analysis and Prediction in Sub-Saharan Africa

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Abstract

Food insecurity continues to be a pressing global challenge, especially in Sub-Saharan Africa where **millions need urgent food assistance**^[1]. To address this crisis, USAID created the Famine Early Warning System Network (FEWS NET), a vital initiative that monitors and forecasts food insecurity. Our project works alongside the UCSB Climate Hazards Center (CHC), which processes satellite data collected by FEWS NET to develop models that **predict crop production and yields**. Timely and precise forecasts are essential for reducing the impacts of climate variability and other shocks to food systems in vulnerable regions. Our capstone project contributes to this larger initiative through three main objectives.

Our objectives include:

- Creating a quality control pipeline for crop yield data in the HarvestStat Dataset
- Developing an application to display and interact with the relevant earth observation and agriculture data
- Developing multiple machine learning models for accurate crop yield forecasting

Introduction

Objective 1: **Reliable estimates of crop yield are crucial** for accurate forecasting and early detection of food insecurity. However, for a multitude of reasons, the data occasionally contains misreported/unrealistic values and has large amounts of missing data. The first objective of our project seeks to address this problem through the **automated detection of data outliers**.

Objective 2: **Visualizing geospatial climate and crop data** is crucial to understanding spatial trends and model performance. When we began this project we struggled to find an effective way of visualizing the data set. This process was often time consuming and required special considerations for each country. The second objective in our project became creating an interactive app that allows quick and easy visualization of crop yields, climate trends, and model performances.

Objective 3: In addition to improving the quality of crop data, our second goal is to leverage this data and Earth Observation data (EO data) to generate **machine learning models to accurately predict future crop yields**. Our team made use of **tree-based** machine learning methods and **clustering** algorithms to produce specialized models for the various subregions in **South Africa** and **Malawi**. The models were trained specifically on these two countries due to the already high quality of their crop and EO data. We also explored the use of a Long Short Term Memory model to understand trends in the data and the relationships between predictors.

Datasets

Our dataset integrates ground-level agricultural records and satellite-based environmental data to support crop yield modeling. It is organized by year, month, and by administrative region (both national and subnational borders)

- Crop Data:** includes crop production, yield, planting and harvest months. Primarily collected from regions in Sub-Saharan Africa such as South Africa.
- Earth Observation (EO) Data:** includes a large number of climate and environmental predictors, such as temperature, precipitation, evaporative demand, various measurements of topsoil moisture
 - Each EO dataset is available in three formats, with and without crop masks. These masks help define agricultural boundaries more precisely, improving the relevance of EO data for crop modeling.

Data Challenges:

- Data Scarcity:** Agricultural data is **inconsistently available** across regions and time periods, with limited records especially in areas with crop failure.
- Data Quality:** Sampling errors, processing limitations, and technical inconsistencies provided additional obstacles to working with the data.
- Bias:** Regions with severe crop losses may not be surveyed or reported, leading to **underrepresentation of critical cases**.

Objective 1: Quality Control

The outliers in our dataset fall into the following two categories:

- Low Variance Outlier:** Consecutive values with extremely low or no variance indicate that the data was misreported/falsified and is potentially not representative of the true crop yield
- Positive Yield Outlier:** Unrealistically Inflated Values

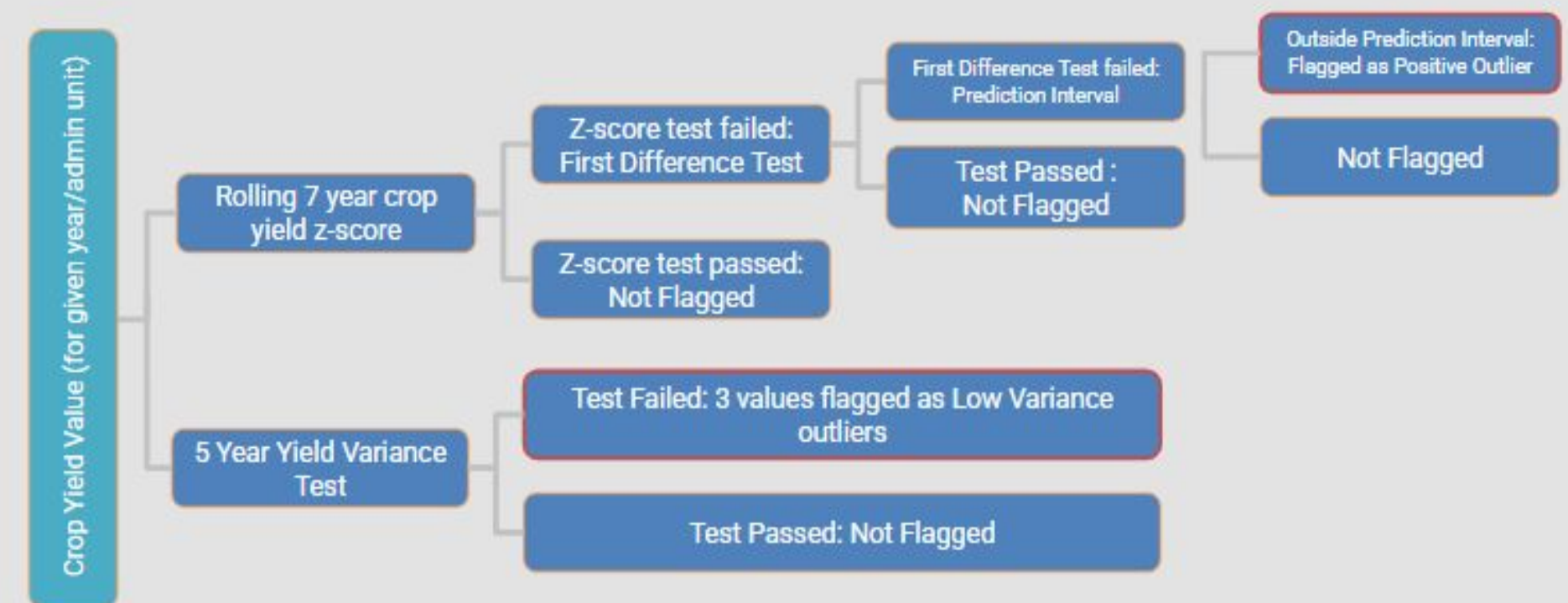
The current method of identifying outliers relies on **manual inspection of time series**. In order to automate this process, we created a decision tree that utilizes multiple tests/methods.

Materials & Methods:

- Rolling 7-year z-score Test:** Mean yield is calculated for rolling intervals of 7 years for each administrative unit. Each observation is given a z-score based on the distribution. If the z-score is greater than 2, the value is given a preliminary flag
- First Difference Test:** Same method as above but using first difference values rather than crop yield
- Prediction Interval:** We built a prediction model that predicts crop yield for an administrative unit given EO feature values. The model utilizes hierarchical clustering to create panels of similar administrative units upon which an epsilon **Support Vector Regression** model is trained. This is used to create a confidence interval the true value is believed to fall in. Values that lie above the upper limit are given a flag
- Yield Variance Test:** Sliding window intervals of 5 years are analyzed for three consecutive crop yields that repeat or don't deviate from a constant linear trend

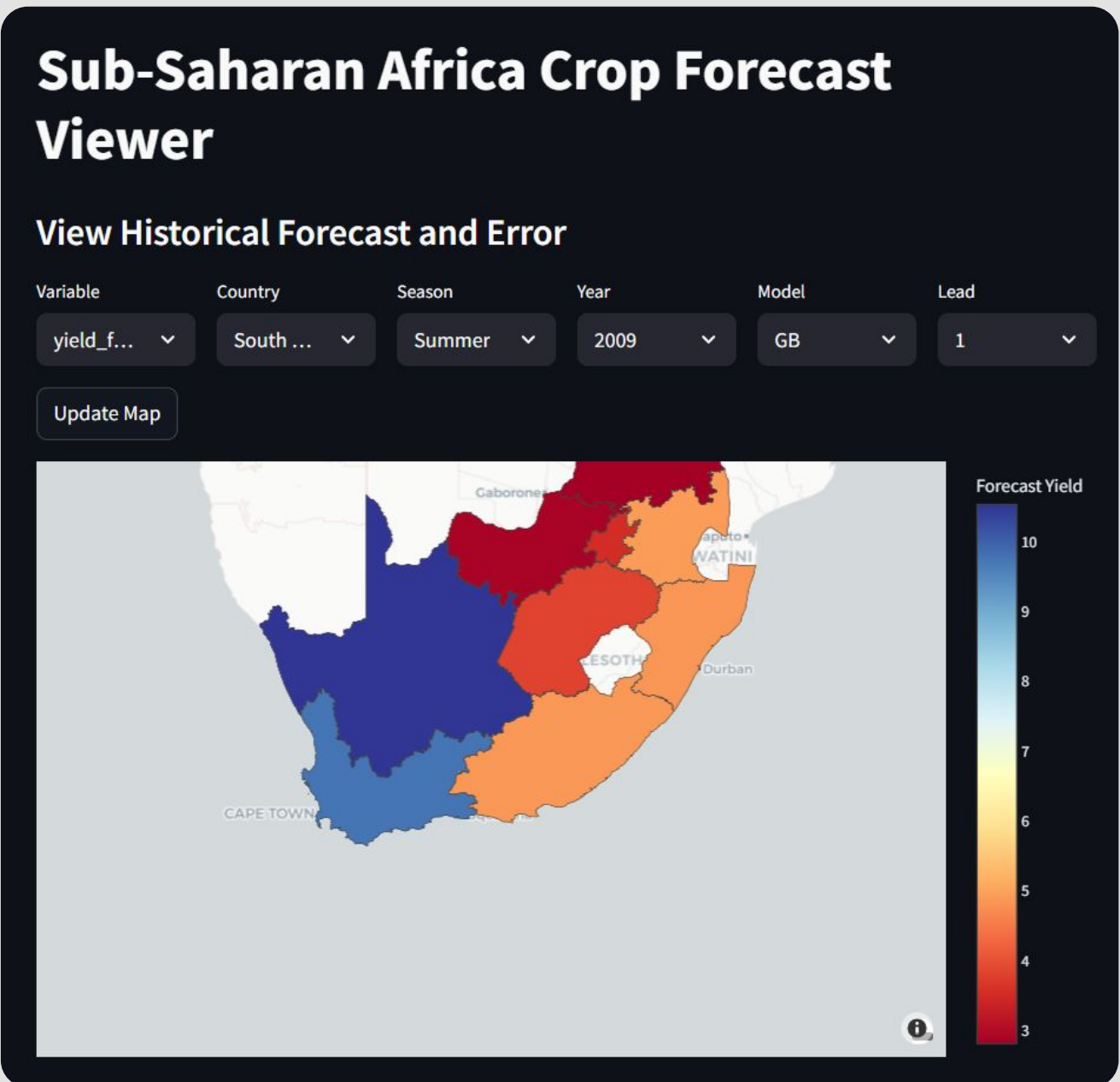
These various methods are combined into the model below that flags crop yield values based on the results.

Outlier Detection Decision Tree Model:



Objective 2: Interactive Visualization App

Working with geospatial data can prove extremely difficult without spatially visualizing the data. Due to the size of our datasets and the difficulty of visualizing the many features, **we created an online dashboard using Streamlit to allow anyone to visualize both critical crop and earth observation data, and historical yield predictions of various models.**

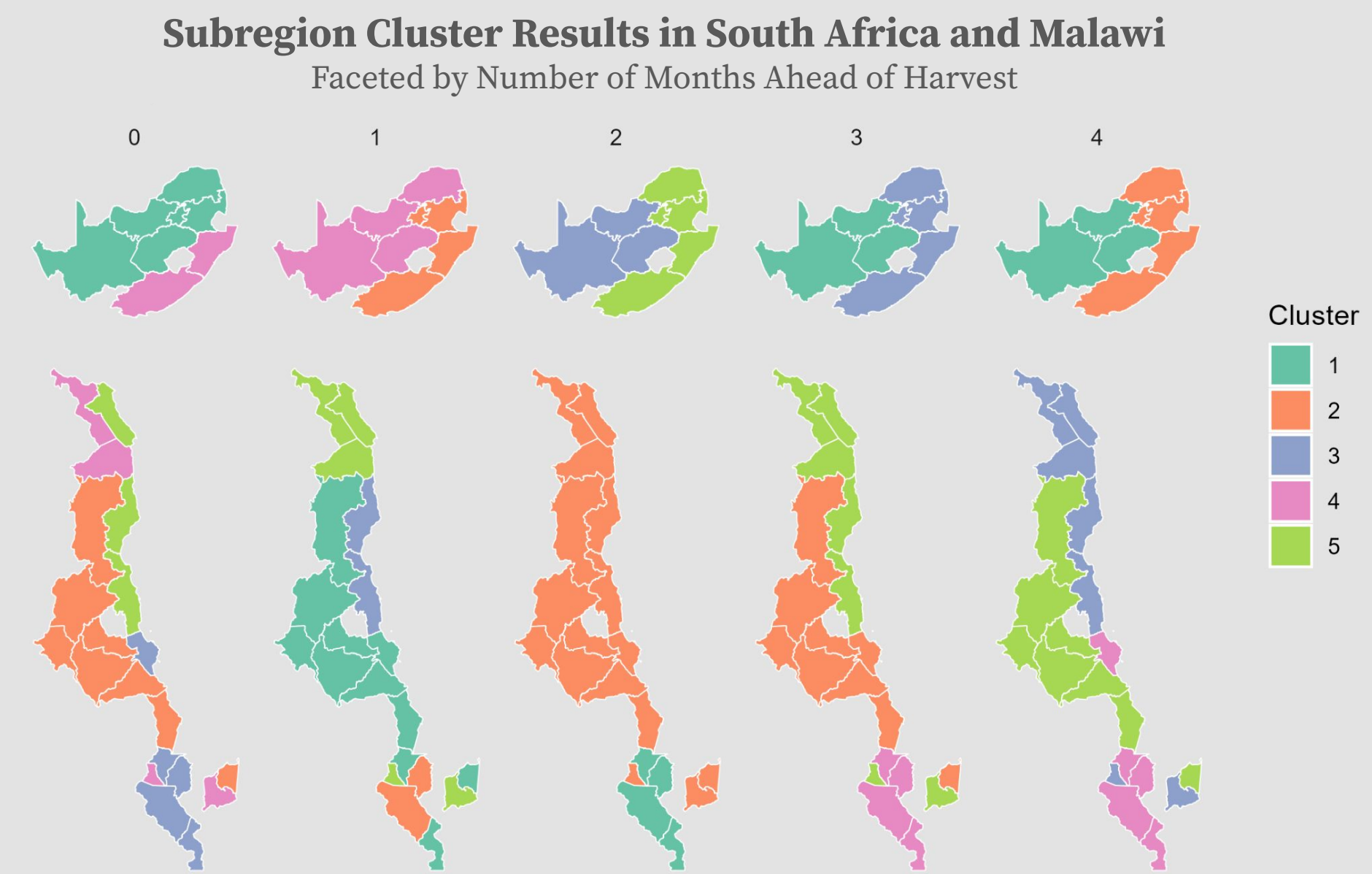


The dashboard is capable of displaying yield predictions generated by a forecasting model, with adjustable filters for season, harvest year, model, and other features.

Objective 3: Crop Yield Forecasting

Model 1: Random Forest + Clustering

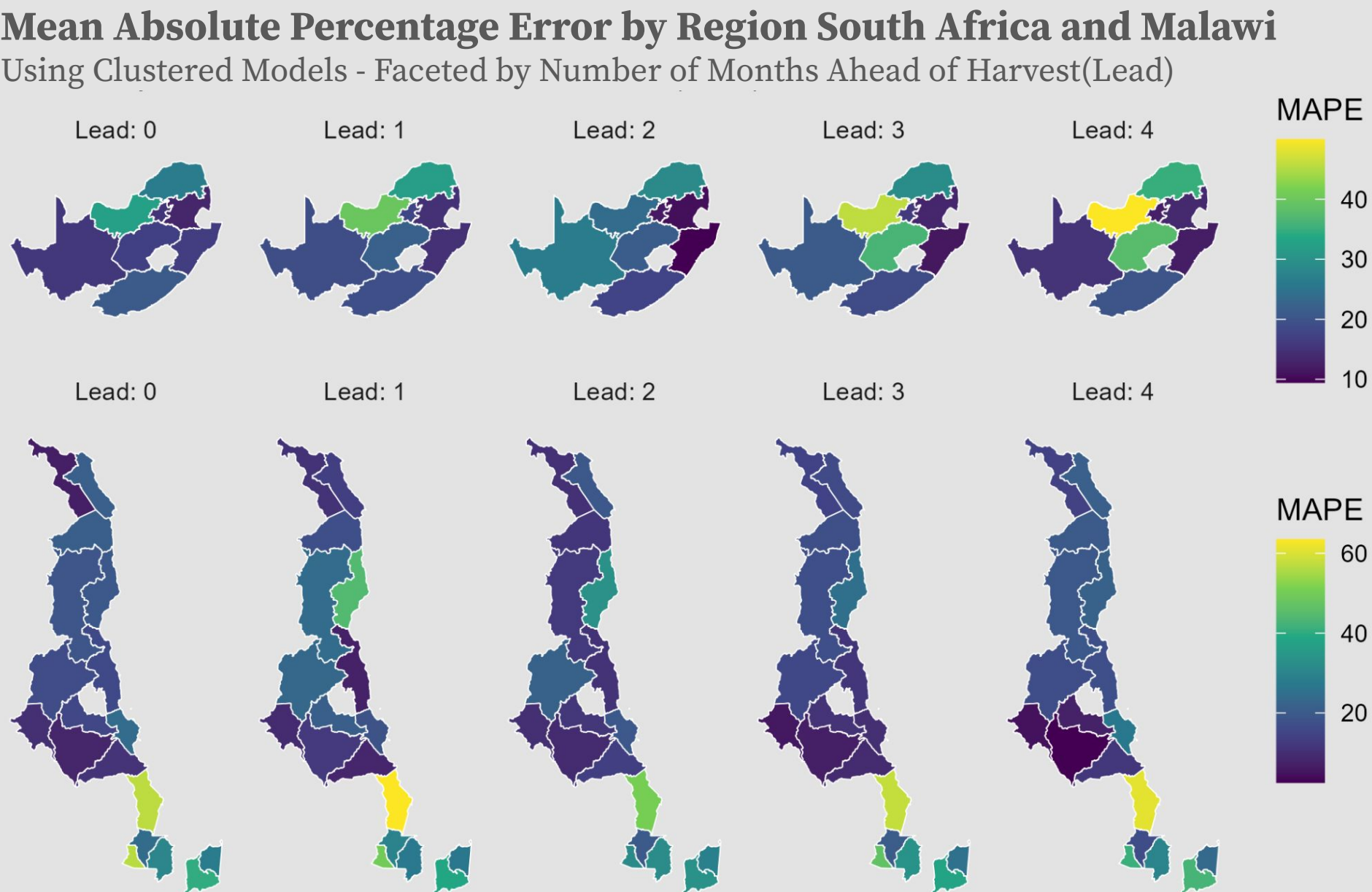
Our group fit **random forest models** and utilized **K-means clustering** on a dataset of earth observation and crop yield data to predict maize yields for similar subregions in Malawi and South Africa. Additionally for each group, we fit separate models each using earth observation data recorded between 0 to 4 months before the harvest date.



After aggregating each model's predictions, we found a **significant reduction in the error** for yields in South Africa through the use of clustering, but did not find a similar performance improvement in Malawi. We also noticed that the models performed poorly on two subregions of Malawi, (*Chikwawa* and *Nsanje*), raising the overall error in Malawi.

Mean Average Percent Error for Model Prediction		
Model Type	South Africa	Malawi
Clustering	21.95	51.41
No Clustering	35.14	42.45
Clustering (Abnormal Zones Removed)	21.95	23.07
No Clustering (Abnormal Zones Removed)	35.14	23.62

This table displays MAPE scores aggregated across all lead times



The above visual for Malawi excludes the regions Chikwawa and Nsanje

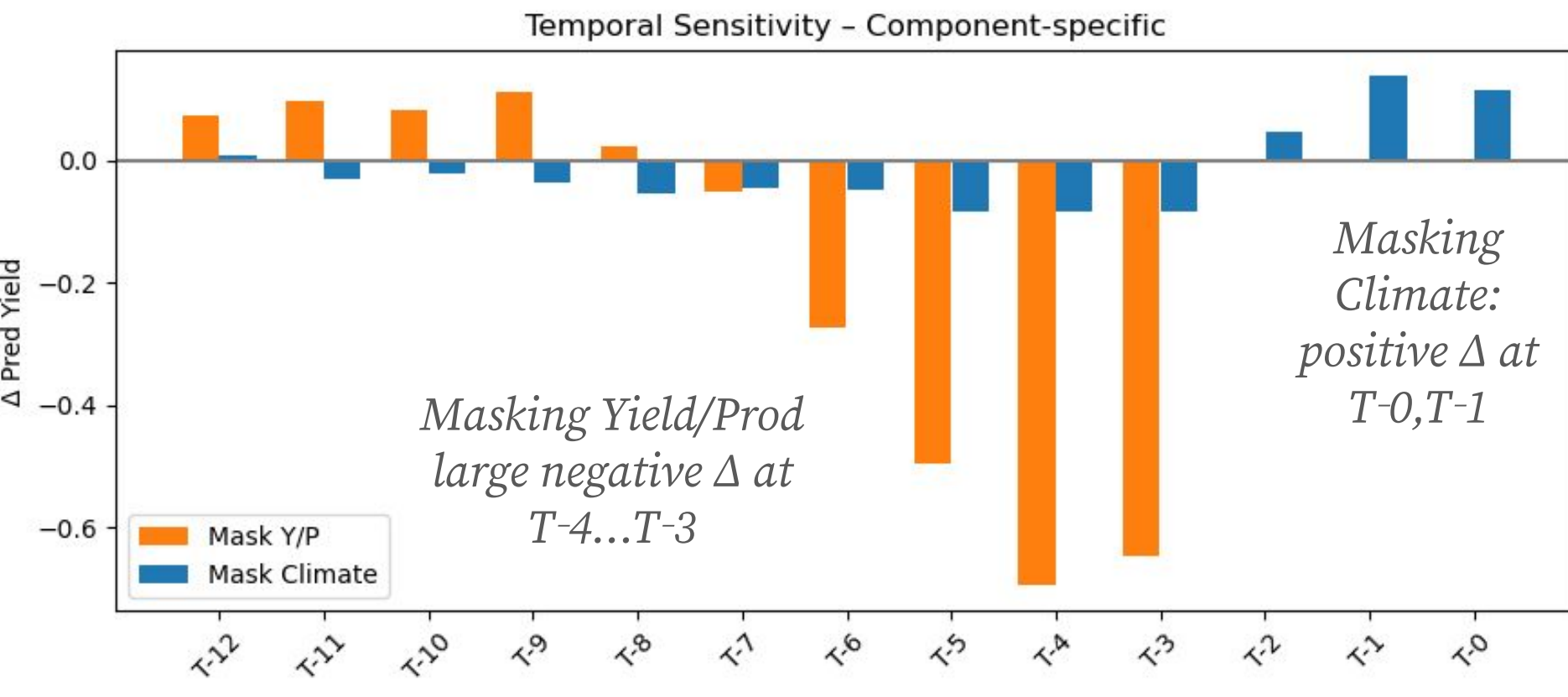
We implemented an **expanding window lagged time series cross validation approach** to train the model to reflect the delay in how crop data is typically collected. This strategy involves splitting the training data to start with a fixed window (e.g., 10 years) that expands one year at a time.

For each fold:

- Train the model on all data from the start year (e.g., 1980) up to a cutoff year (e.g., 1989).
- Skip the next year (e.g., 1990) to account for the delay in data availability.
- Validate on the following year (e.g., 1991), simulating a real-world forecast where recent data isn't available at prediction time.
- Repeat the process, expanding the training window and moving the validation year forward.

Model 2: LSTM

We trained a **sequence-to-one LSTM** on 13-year windows (T-12 → T) and found that masking Yield/Production at T-4 & T-3 led to a sharp drop in the predicted yield and masking climate features at T-0 & T-1 led to a slight increase.



T-12...T-3: all predictors (yield, production & 6 EO climate variables)
T-2...T: only climate variables (yield/production masked & detached).

Findings and Future Work

Our group was able to make progress in our three objectives of improving data quality, producing a dashboard for visualizing crop data, and predicting crop yields through machine-learning models.

Objective 1: We were able to develop an outlier detection model capable of identifying suspect crop yield values while **avoiding manual inspection** of the time series. In the future we hope to further investigate methods of **imputing missing or erroneous data** and incorporate the results of the outlier detection process into our predictive modeling.

Objective 2: We developed and **hosted an interactive web app** for visualizing observed and predicted crop yield as well as the environmental features informing our predictive models. This app may be further developed by allowing for the visualization of our **predictive models' forecasts** and residuals. We may also provide a tab for visualizing the amount of **outliers or missingness** in the data with user-defined thresholds for determining what is an outlier.

Objective 3: We explored several machine learning models' effectiveness in predicting crop yield from the environmental features provided to us, ultimately deciding to focus on **Random Forest** and **LSTM** models. We also performed **K-Means clustering** and fit a Random Forest for each cluster which led to a **drop in the MAPE scores for South Africa**. Looking ahead, we'd like to improve our model performance for Malawi and expand our scope to other countries. We also may want to pursue methods of leveraging the spatial correlations in the data in order to use one country's data to forecast crop yield in another country with potentially lower data availability. We may also evaluate the South Africa-trained LSTM's generalization ability by applying it directly to Malawi maize data and quantifying zero-shot transfer performance

References and Acknowledgments

[1] Lee, Davenport, Shukla, Husak, Funk, Harrison, McNally, Rowland, Budde, Verdin (2022). *Maize yield forecasts for Sub-Saharan Africa using Earth Observation data and machine learning*. Global Food Security, Volume 33.

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